MERIDIAN FREIGHT

Dock staffing before the rush

Dock visibility and labor planning
A 30-day Memphis pilot

Fictional carrier. Synthetic demonstration.



The loading dock sets the pace

Meridian Freight transfers loads between trailers across three cross-dock terminals.

A trailer needs an available door and a crew. When either falls behind, the next load waits.

36

loading doors

Atlanta	12
Dallas	10
Memphis	**14**

The manager's decision

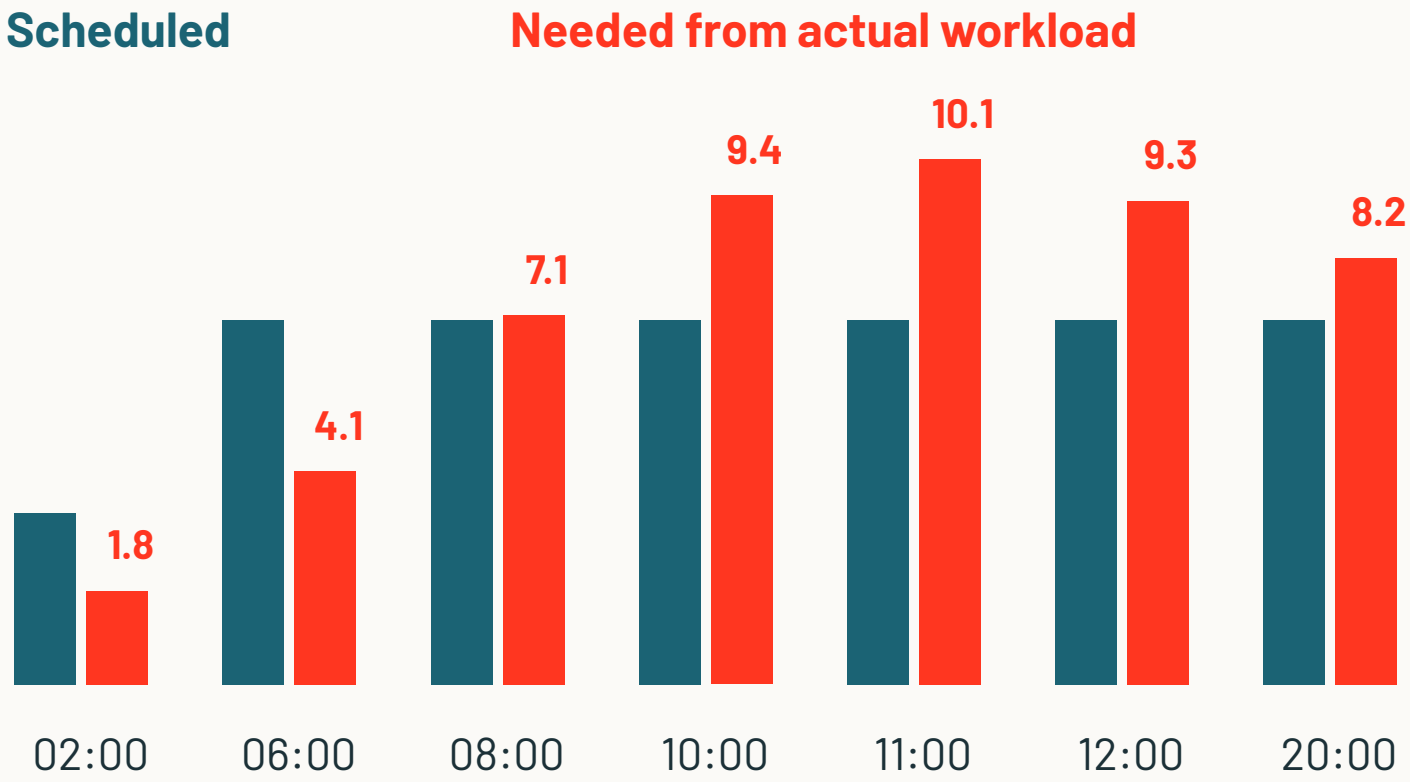
Which doors and workers need attention before the next arrival wave?

Fictional carrier. Synthetic demonstration.

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The rush arrives. The roster stays flat

Average workers per terminal, selected hours



11:00

10.1

workers needed

7.0 scheduled

3.1 worker gap

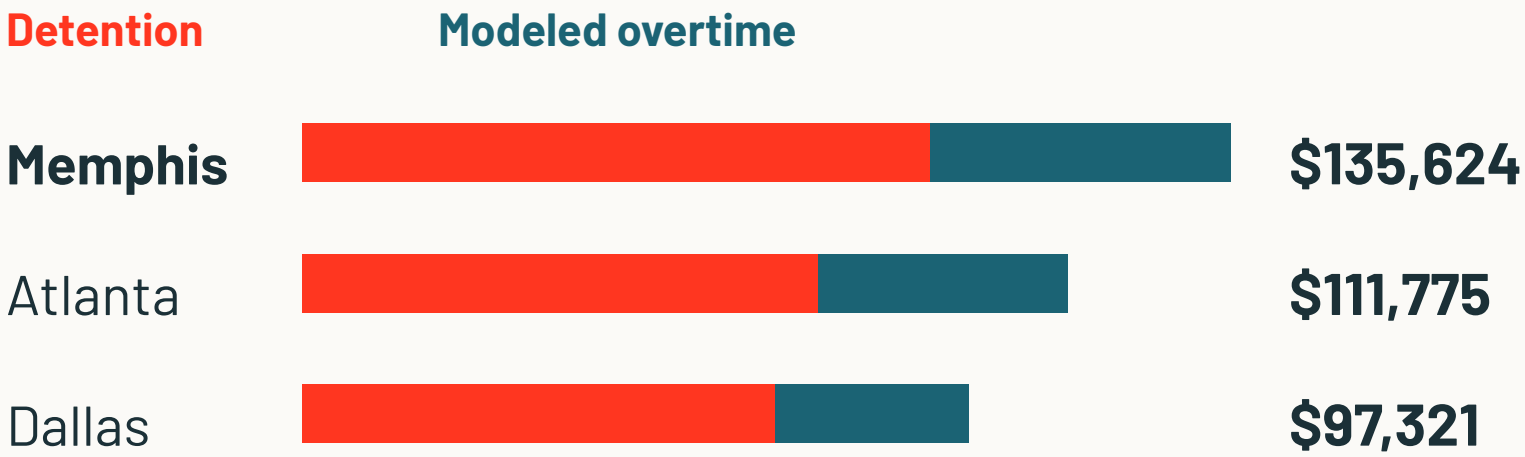
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The mismatch creates a measurable cost

$344,720

combined cost across 31 synthetic days



Why Memphis first

$91,650

highest detention cost

Start with one terminal.
Measure before scaling.

Fictional carrier. Synthetic demonstration.

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A different start to the next shift

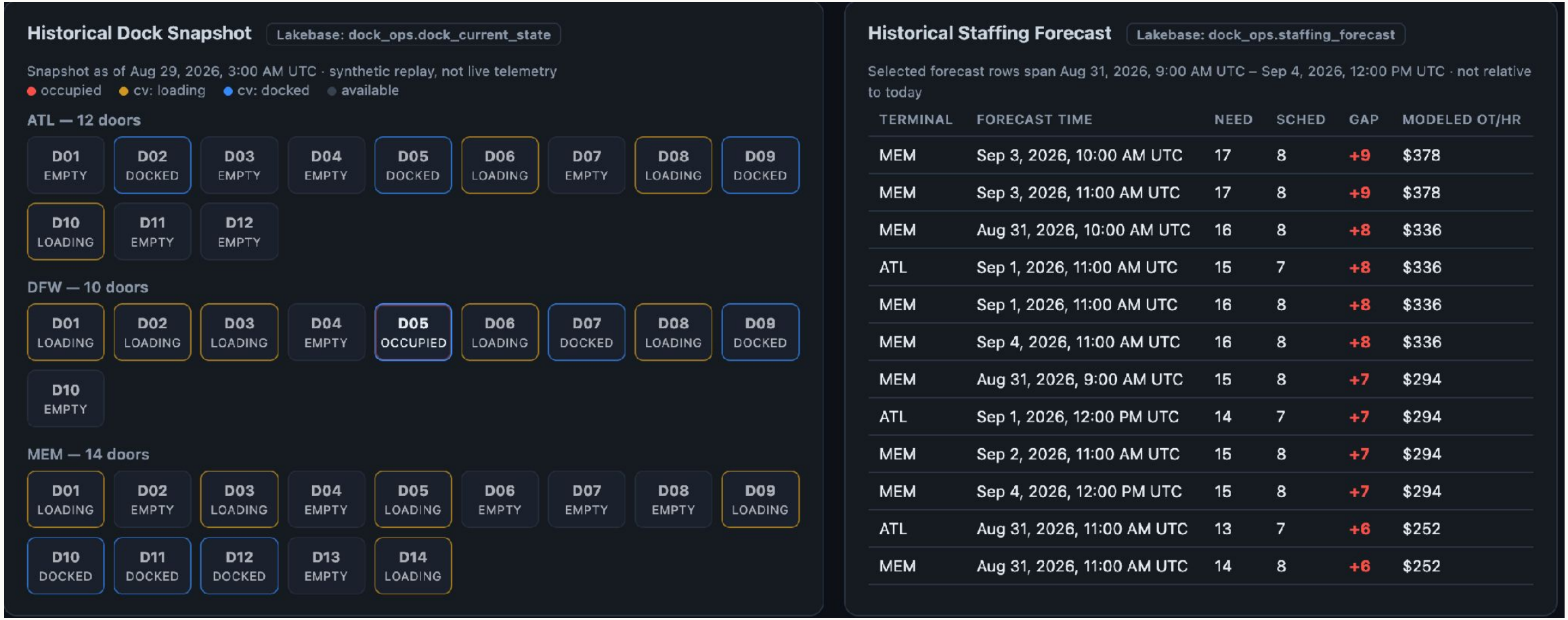
Current pattern	Proposed operating routine
Review yesterday's report	**Check the latest dock snapshot**
Carry forward the fixed roster	**Review the dated staffing gaps**
Add overtime after queues grow	**Approve staffing before the rush**

The supervisor approves the change and records the result.

Fictional carrier. Synthetic demonstration.

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One operating view for all 36 doors



TERMINAL	FORECAST TIME	NEED	SCHED	GAP	MODELED OT/HR
MEM	Sep 3, 2026, 10:00 AM UTC	17	8	+9	$378
MEM	Sep 3, 2026, 11:00 AM UTC	17	8	+9	$378
MEM	Aug 31, 2026, 10:00 AM UTC	16	8	+8	$336
ATL	Sep 1, 2026, 11:00 AM UTC	15	7	+8	$336
MEM	Sep 1, 2026, 11:00 AM UTC	16	8	+8	$336
MEM	Sep 4, 2026, 11:00 AM UTC	16	8	+8	$336
MEM	Aug 31, 2026, 9:00 AM UTC	15	8	+7	$294
ATL	Sep 1, 2026, 12:00 PM UTC	14	7	+7	$294
MEM	Sep 2, 2026, 11:00 AM UTC	15	8	+7	$294
MEM	Sep 4, 2026, 12:00 PM UTC	15	8	+7	$294
ATL	Aug 31, 2026, 11:00 AM UTC	13	7	+6	$252
MEM	Aug 31, 2026, 11:00 AM UTC	14	8	+6	$252

Open the live console

Fictional carrier. Synthetic demonstration.

Follow the Memphis staffing gap

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A recommendation the supervisor can act on

MEMPHIS 3 SEP 2026 10:00 UTC

+9

additional workers

17 required
8 scheduled

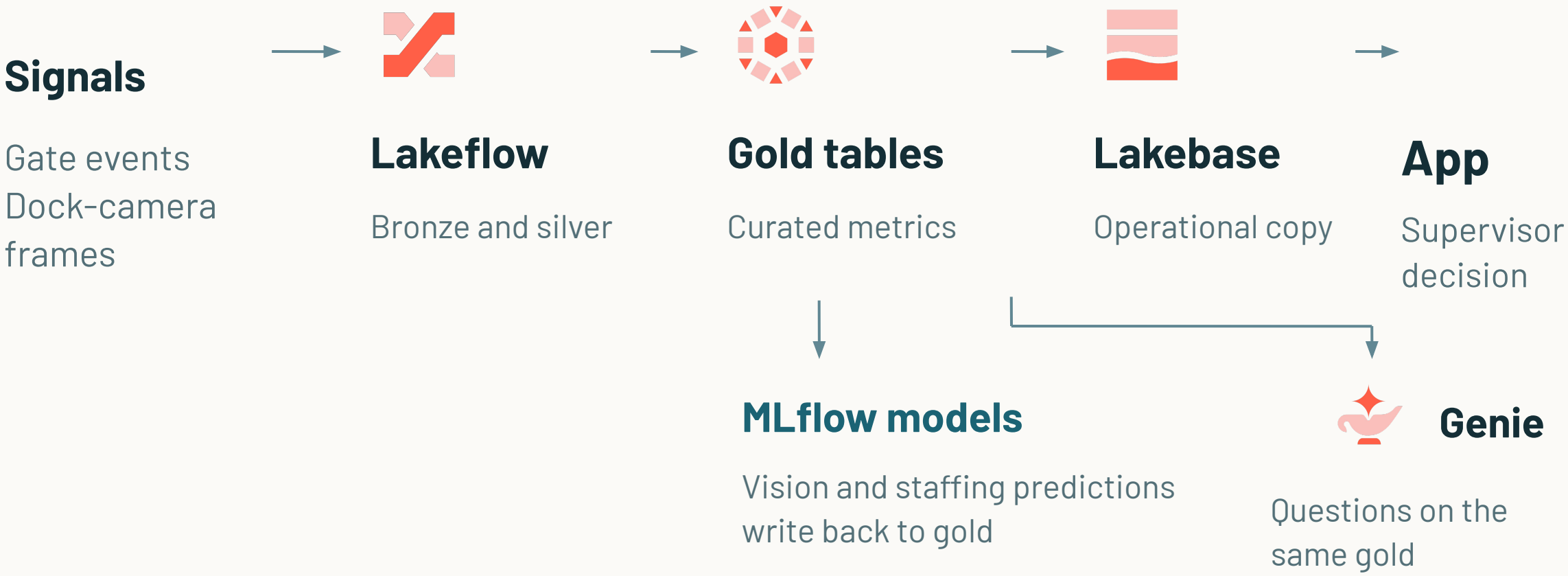
Supervisor checks availability, qualifications, and shift rules before approving a move.

Open the briefing

Ask a leadership follow-up in Genie

Fictional carrier. Synthetic demonstration.

The data journey behind that decision



Unity Catalog governs lakehouse data and models. Postgres uses database roles.

Fictional carrier. Synthetic demonstration.

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Prototype evidence and pilot success measures

Verified prototype	Proposed pilot acceptance
36 doors served through Lakebase	**At least 10% lower detention per trailer**
Forecast and briefing connected	**No increase in overtime per trailer**
Genie answer reconciled to source	**Forecast MAE at or below 2 workers**
Four synthetic camera states classified	**p95 data age at or below 5 minutes**

Agree targets after baseline review. Test them with real pilot feeds and matched shifts.

Fictional carrier. Synthetic demonstration.

A 30-day pilot with owners and decision gates

Scope: Memphis, existing feeds, supervisor-approved staffing changes

When	Deliverable	Accountable owner	Decision gate
Week 1	Feeds connected Baseline scorecard	Platform architect Terminal manager	Data ready and baseline agreed
Week 2	Shadow staffing plan Camera validation	Analytics lead Terminal manager	Forecast and freshness accepted
Week 3	Controlled staffing trial Action log	Terminal manager	Supervisor and safety approval
Week 4	Matched-shift results Net cost assessment	VP Operations Finance partner	Scale, revise, or stop

Clock starts after feed access and pilot ownership are confirmed.

Fictional carrier. Synthetic demonstration.

The Memphis business case

31-day synthetic cost baseline

$135,624

$91,650 detention
$43,974 modeled overtime

At the proposed 10% detention target

$9,165

gross benefit per comparable 31 days

Net benefit must deduct incremental labor, platform, and implementation costs.

Illustrative sensitivity only. Replace synthetic costs with the agreed pilot baseline.

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Decision today: a Memphis pilot

Authorize one terminal and a measured trial.

Within 2 business days

VP Operations

Name the Memphis sponsor and terminal owner

Within 5 business days

Terminal manager + architect

Confirm feeds, access, and staffing constraints

At a 45-minute kickoff

Sponsor + finance + delivery

Approve scorecard, cost cap, and trial start

Kickoff output: signed pilot scope, named owners, and an agreed success scorecard

Fictional carrier. Synthetic demonstration.

Appendix: technical evidence and limits

Component	Evidence	Pilot validation still required
Staffing model	MAE 1.885 workers 5.6% better than baseline	Real-demand holdout Staffing and skills constraints
Camera classifier	Four states verified live 100% synthetic holdout	Real cameras, lighting, occlusion, and drift
Serving and governance	App reads Lakebase Worker-ID column mask in UC	Continuous sync and freshness Separate Postgres grants
AI briefing and Genie	Dated recommendations SQL and source rows inspectable	Human review and identity validation at each boundary

Forecast metrics describe the original synthetic training snapshot.

Fictional carrier. Synthetic demonstration.

Appendix: presenter navigation

Operations console	Dock snapshot, dated staffing plan, and briefing
Genie	Ask about terminal cost, then inspect SQL
Workspace walkthrough	Source reconciliation, forecasts, and governance
Successful validation run	Read-only walkthrough and reconciled totals

19 minutes for the story and demo. 6 minutes for questions.

Fictional carrier. Synthetic demonstration.

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